

WebHDFS REST API

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Document Conventions

Monospaced	Used for commands, HTTP request and responses and code blocks.
<code><Monospaced></code>	User entered values.
<code>[Monospaced]</code>	Optional values. When the value is not specified, the default value is used.
<i>Italics</i>	Important phrases and words.

Introduction

The HTTP REST API supports the complete `FileSystem/FileContext` interface for HDFS. The operations and the corresponding `FileSystem/FileContext` methods are shown in the next section. The Section `HTTP Query Parameter Dictionary` specifies the parameter details such as the defaults and the valid values.

Operations

- HTTP GET
 - OPEN (see `FileSystem.open`)
 - GETFILESTATUS (see `FileSystem.getFileStatus`)
 - LISTSTATUS (see `FileSystem.listStatus`)
 - GETCONTENTSUMMARY (see `FileSystem.getContentSummary`)
 - GETFILECHECKSUM (see `FileSystem.getFileChecksum`)
 - GETHOMEDIRECTORY (see `FileSystem.getHomeDirectory`)
 - GETDELEGATIONTOKEN (see `FileSystem.getDelegationToken`)
 - GETDELEGATIONTOKENS (see `FileSystem.getDelegationTokens`)
 - GETXATTRS (see `FileSystem.getXAttr`)
 - GETXATTRS (see `FileSystem.getXAttrs`)
 - GETXATTRS (see `FileSystem.getXAttrs`)
 - LISTXATTRS (see `FileSystem.listXAttrs`)
 - CHECKACCESS (see `FileSystem.access`)
- HTTP PUT
 - CREATE (see `FileSystem.create`)
 - MKDIRS (see `FileSystem.mkdirs`)
 - CREATESYMLINK (see `FileContext.createSymlink`)
 - RENAME (see `FileSystem.rename`)
 - SETREPLICATION (see `FileSystem.setReplication`)
 - SETOWNER (see `FileSystem.setOwner`)
 - SETPERMISSION (see `FileSystem.setPermission`)
 - SETTIMES (see `FileSystem.setTimes`)
 - RENEWDELEGATIONTOKEN (see `FileSystem.renewDelegationToken`)
 - CANCELDELEGATIONTOKEN (see `FileSystem.cancelDelegationToken`)
 - CREATESNAPSHOT (see `FileSystem.createSnapshot`)
 - RENAMESNAPSHOT (see `FileSystem.renameSnapshot`)
 - SETXATTR (see `FileSystem.setXAttr`)
 - REMOVEXATTR (see `FileSystem.removeXAttr`)
- HTTP POST
 - APPEND (see `FileSystem.append`)
 - CONCAT (see `FileSystem.concat`)
 - TRUNCATE (see `FileSystem.concat`)
- HTTP DELETE
 - DELETE (see `FileSystem.delete`)
 - DELETESNAPSHOT (see `FileSystem.deleteSnapshot`)

FileSystem URIs vs HTTP URLs

The FileSystem scheme of WebHDFS is "webhdfs://". A WebHDFS FileSystem URI has the following format.

```
webhdfs://<HOST>:<HTTP_PORT>/<PATH>
```

The above WebHDFS URI corresponds to the below HDFS URI.

```
hdfs://<HOST>:<RPC_PORT>/<PATH>
```

In the REST API, the prefix "/webhdfs/v1" is inserted in the path and a query is appended at the end. Therefore, the corresponding HTTP URL has the following format.

```
http://<HOST>:<HTTP_PORT>/webhdfs/v1/<PATH>?op=...
```

HDFS Configuration Options

Below are the HDFS configuration options for WebHDFS.

Property Name	Description
dfs.webhdfs.enabled	Enable/disable WebHDFS in Namenodes and Datanodes
dfs.web.authentication.kerberos.principal	The HTTP Kerberos principal used by Hadoop-Auth in the HTTP endpoint. The HTTP Kerberos principal MUST start with 'HTTP/' per Kerberos HTTP SPNEGO specification. A value of "*" will use all HTTP principals found in the keytab.
dfs.web.authentication.kerberos.keytab	The Kerberos keytab file with the credentials for the HTTP Kerberos principal used by Hadoop-Auth in the HTTP endpoint.

Authentication

When security is *off*, the authenticated user is the username specified in the `user.name` query parameter. If the `user.name` parameter is not set, the server may either set the authenticated user to a default web user, if there is any, or return an error response.

When security is *on*, authentication is performed by either Hadoop delegation token or Kerberos SPNEGO. If a token is set in the `delegation` query parameter, the authenticated user is the user encoded in the token. If the `delegation` parameter is not set, the user is authenticated by Kerberos SPNEGO.

Below are examples using the `curl` command tool.

1. Authentication when security is off:

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?[user.name=<USER>&]op=..."
```

2. Authentication using Kerberos SPNEGO when security is on:

```
curl -i --negotiate -u : "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=..."
```

3. Authentication using Hadoop delegation token when security is on:

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?delegation=<TOKEN>&op=..."
```

See also: [Authentication for Hadoop HTTP web-consoles](#)

Proxy Users

When the proxy user feature is enabled, a proxy user P may submit a request on behalf of another user U . The username of U must be specified in the doas query parameter unless a delegation token is presented in authentication. In such case, the information of both users P and U must be encoded in the delegation token.

1. A proxy request when security is off:

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?[user.name=<USER>]&doas=<USER>&op=...
```

2. A proxy request using Kerberos SPNEGO when security is on:

```
curl -i --negotiate -u : "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?doas=<USER>&op=..."
```

3. A proxy request using Hadoop delegation token when security is on:

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?delegation=<TOKEN>&op=..."
```

File and Directory Operations

Create and Write to a File

- Step 1: Submit a HTTP PUT request without automatically following redirects and without sending the file data.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=CREATE
[&overwrite=<true | false>][&blocksize=<LONG>][&replication=<SHORT>]
[&permission=<OCTAL>][&buffersize=<INT>]"
```

The request is redirected to a datanode where the file data is to be written:

```
HTTP/1.1 307 TEMPORARY_REDIRECT
Location: http://<DATANODE>:<PORT>/webhdfs/v1/<PATH>?op=CREATE...
Content-Length: 0
```

- Step 2: Submit another HTTP PUT request using the URL in the Location header with the file data to be written.

```
curl -i -X PUT -T <LOCAL_FILE> "http://<DATANODE>:<PORT>/webhdfs/v1/<PATH>?op=CREAT
```

The client receives a 201 Created response with zero content length and the WebHDFS URI of the file in the Location header:

```
HTTP/1.1 201 Created
Location: webhdfs://<HOST>:<PORT>/<PATH>
Content-Length: 0
```

Note that the reason of having two-step create/append is for preventing clients to send out data before the redirect. This issue is addressed by the "Expect: 100-continue" header in HTTP/1.1; see [RFC 2616, Section 8.2.3](#). Unfortunately, there are software library bugs (e.g. Jetty 6 HTTP server and Java 6 HTTP client), which do not correctly implement "Expect: 100-continue". The two-step create/append is a temporary workaround for the software library bugs.

See also: [overwrite](#), [blocksize](#), [replication](#), [permission](#), [buffer size](#), [FileSystem.create](#)

Append to a File

- Step 1: Submit a HTTP POST request without automatically following redirects and without sending the file data.

```
curl -i -X POST "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=APPEND[&buffer size=<INT>
```

The request is redirected to a datanode where the file data is to be appended:

```
HTTP/1.1 307 TEMPORARY_REDIRECT
Location: http://<DATANODE>:<PORT>/webhdfs/v1/<PATH>?op=APPEND...
Content-Length: 0
```

- Step 2: Submit another HTTP POST request using the URL in the Location header with the file data to be appended.

```
curl -i -X POST -T <LOCAL_FILE> "http://<DATANODE>:<PORT>/webhdfs/v1/<PATH>?op=APPE
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See the note in the previous section for the description of why this operation requires two steps.

See also: [bufferSize](#), [FileSystem.append](#)

Concat File(s)

- Submit a HTTP POST request.

```
curl -i -X POST "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=CONCAT&sources=<PATHS>"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [sources](#), [FileSystem.concat](#)

Open and Read a File

- Submit a HTTP GET request with automatically following redirects.

```
curl -i -L "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=OPEN
[&offset=<LONG>][&length=<LONG>][&bufferSize=<INT>]"
```

The request is redirected to a datanode where the file data can be read:

```
HTTP/1.1 307 TEMPORARY_REDIRECT
Location: http://<DATANODE>:<PORT>/webhdfs/v1/<PATH>?op=OPEN...
Content-Length: 0
```

The client follows the redirect to the datanode and receives the file data:

```
HTTP/1.1 200 OK
Content-Type: application/octet-stream
Content-Length: 22

Hello, webhdfs user!
```

See also: [offset](#), [length](#), [bufferSize](#), [FileSystem.open](#)

Make a Directory

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=MKDIRS[&permission=<OCTAL
```

The client receives a response with a [boolean JSON object](#):

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{"boolean": true}
```

See also: [permission](#), [FileSystem.mkdirs](#)

Create a Symbolic Link

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=CREATE_SYMLINK
                &destination=<PATH>[&createParent=<true | false>]"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [destination](#), [createParent](#), [FileSystem.createSymlink](#)

Rename a File/Directory

- Submit a HTTP PUT request.

```
curl -i -X PUT "<HOST>:<PORT>/webhdfs/v1/<PATH>?op=RENAME&destination=<PATH>"
```

The client receives a response with a [boolean JSON object](#):

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{"boolean": true}
```

See also: [destination](#), [FileSystem.rename](#)

Delete a File/Directory

- Submit a HTTP DELETE request.

```
curl -i -X DELETE "http://<host>:<port>/webhdfs/v1/<path>?op=DELETE
[&recursive=<true |false>]"
```

The client receives a response with a [boolean JSON object](#):

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{"boolean": true}
```

See also: [recursive](#), [FileSystem.delete](#)

Truncate a File

- Submit a HTTP POST request.

```
curl -i -X POST "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=TRUNCATE&newlength=<LONG>"
```



The client receives a response with a `boolean` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{"boolean": true}
```

See also: `newlength`, `FileSystem.truncate`

Status of a File/Directory

- Submit a HTTP GET request.

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=GETFILESTATUS"
```

The client receives a response with a `FileStatus` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "FileStatus":
  {
    "accessTime"      : 0,
    "blockSize"       : 0,
    "group"           : "supergroup",
    "length"          : 0,           //in bytes, zero for directories
    "modificationTime": 1320173277227,
    "owner"           : "webuser",
    "pathSuffix"       : "",
    "permission"       : "777",
    "replication"      : 0,
    "type"            : "DIRECTORY" //enum {FILE, DIRECTORY, SYMLINK}
  }
}
```

See also: `FileSystem.getFileStatus`

List a Directory

- Submit a HTTP GET request.

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=LISTSTATUS"
```

The client receives a response with a `FileStatuses` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 427

{
  "FileStatuses":
  {
    "FileStatus":
    [
      {
        "accessTime"      : 1320171722771,
        "blockSize"       : 33554432,
        "group"           : "supergroup",
        "length"          : 24930,
        "modificationTime": 1320171722771,
        "owner"           : "webuser",
        "pathSuffix"       : "a.patch",
        "permission"       : "644",
        "replication"      : 1,
        "type"            : "FILE"
      },
      {
        "accessTime"      : 0,
        "blockSize"       : 0,
        "group"           : "supergroup",
        "length"          : 0,
        "modificationTime": 1320895981256,
        "owner"           : "szetszwo",
        "pathSuffix"       : "bar",
        "permission"       : "711",
        "replication"      : 0,
        "type"            : "DIRECTORY"
      },
      ...
    ]
  }
}
```

See also: `FileSystem.listStatus`

Other File System Operations

Get Content Summary of a Directory

- Submit a HTTP GET request.

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=GETCONTENTSUMMARY"
```

The client receives a response with a [ContentSummary](#) JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "ContentSummary":
  {
    "directoryCount": 2,
    "fileCount"      : 1,
    "length"         : 24930,
    "quota"           : -1,
    "spaceConsumed"  : 24930,
    "spaceQuota"     : -1
  }
}
```

See also: [FileSystem.getContentSummary](#)

Get File Checksum

- Submit a HTTP GET request.

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=GETFILECHECKSUM"
```

The request is redirected to a datanode:

```
HTTP/1.1 307 TEMPORARY_REDIRECT
Location: http://<DATANODE>:<PORT>/webhdfs/v1/<PATH>?op=GETFILECHECKSUM...
Content-Length: 0
```

The client follows the redirect to the datanode and receives a [FileChecksum](#) JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "FileChecksum":
  {
    "algorithm": "MD5-of-1MD5-of-512CRC32",
    "bytes"     : "eadb10de24aa315748930df6e185c0d ...",
    "length"    : 28
  }
}
```

See also: [FileSystem.getFileChecksum](#)

Get Home Directory

- Submit a HTTP GET request.

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/?op=GETHOMEDIRECTORY"
```

The client receives a response with a [Path JSON object](#):

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{"Path": "/user/szetszwo"}
```

See also: [FileSystem.getHomeDirectory](#)

Set Permission

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=SETPERMISSION
[&permission=<OCTAL>]"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [permission](#), [FileSystem.setPermission](#)

Set Owner

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=SETOWNER
[&owner=<USER>][&group=<GROUP>]"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [owner](#), [group](#), [FileSystem.setOwner](#)

Set Replication Factor

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=SETREPLICATION
[&replication=<SHORT>]"
```

The client receives a response with a [boolean JSON object](#):

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{"boolean": true}
```

See also: [replication](#), [FileSystem.setReplication](#)

Set Access or Modification Time

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=SETTIMES
[&modificationtime=<TIME>][&accesstime=<TIME>]"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [modificationtime](#), [accesstime](#), [FileSystem.setTimes](#)

Modify ACL Entries

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=MODIFYACLENTRIES  
&aclspec=<ACLSPEC>"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK  
Content-Length: 0
```

See also: [FileSystem.modifyAclEntries](#)

Remove ACL Entries

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=REMOVEACLENTRIES  
&aclspec=<ACLSPEC>"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK  
Content-Length: 0
```

See also: [FileSystem.removeAclEntries](#)

Remove Default ACL

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=REMOVEDEFAULTACL"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [FileSystem.removeDefaultAcl](#)

Remove ACL

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=REMOVEACL"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [FileSystem.removeAcl](#)

Set ACL

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=SETACL
               &aclspec=<ACLSPEC>"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [FileSystem.setAcl](#)

Get ACL Status

- Submit a HTTP GET request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=GETACLSTATUS"
```

The client receives a response with a `AclStatus` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "AclStatus": {
    "entries": [
      "user:carla:rw-",
      "group::r-x"
    ],
    "group": "supergroup",
    "owner": "hadoop",
    "permission": "775",
    "stickyBit": false
  }
}
```

See also: `FileSystem.getAclStatus`

Check access

- Submit a HTTP GET request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=CHECKACCESS
               &fsaction=<FSACTION>"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: `FileSystem.access`

Extended Attributes(XAttrs) Operations

Set XAttr

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=SETXATTR
                &xattr.name=<XATTRNAME>&xattr.value=<XATTRVALUE>
                &flag=<FLAG>"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [FileSystem.setXAttr](#)

Remove XAttr

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=REMOVEXATTR
                &xattr.name=<XATTRNAME>"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [FileSystem.removeXAttr](#)

Get an XAttr

- Submit a HTTP GET request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=GETXATTRS
                &xattr.name=<XATTRNAME>&encoding=<ENCODING>"
```

The client receives a response with a `XAttrs` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "XAttrs": [
    {
      "name": "XATTRNAME",
      "value": "XATTRVALUE"
    }
  ]
}
```

See also: `FileSystem.getXAttr`

Get multiple XAttrs

- Submit a HTTP GET request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=GETXATTRS
                &xattr.name=<XATTRNAME1>&xattr.name=<XATTRNAME2>
                &encoding=<ENCODING>"
```

The client receives a response with a `XAttrs` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "XAttrs": [
    {
      "name": "XATTRNAME1",
      "value": "XATTRVALUE1"
    },
    {
      "name": "XATTRNAME2",
      "value": "XATTRVALUE2"
    }
  ]
}
```

See also: `FileSystem.getXAttrs`

Get all XAttrs

- Submit a HTTP GET request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=GETXATTRS
               &encoding=<ENCODING>"
```

The client receives a response with a `XAttrs` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "XAttrs": [
    {
      "name": "XATTRNAME1",
      "value": "XATTRVALUE1"
    },
    {
      "name": "XATTRNAME2",
      "value": "XATTRVALUE2"
    },
    {
      "name": "XATTRNAME3",
      "value": "XATTRVALUE3"
    }
  ]
}
```

See also: [FileSystem.getXAttrs](#)

List all XAttrs

- Submit a HTTP GET request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=LISTXATTRS"
```

The client receives a response with a `XAttrNames` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "XAttrNames": "[\"XATTRNAME1\", \"XATTRNAME2\", \"XATTRNAME3\"]"
}
```

See also: [FileSystem.listXAttrs](#)

Snapshot Operations

Create Snapshot

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=CREATE_SNAPSHOT[&snapshotn
```

The client receives a response with a `Path` JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{"Path": "/user/szetszwo/.snapshot/s1"}
```

See also: [FileSystem.createSnapshot](#)

Delete Snapshot

- Submit a HTTP DELETE request.

```
curl -i -X DELETE "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=DELETE_SNAPSHOT&snapshotn
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [FileSystem.deleteSnapshot](#)

Rename Snapshot

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/<PATH>?op=RENAME_SNAPSHOT
&oldSnapshotName=<SNAPSHOTNAME>&snapshotName=<SNAPSHOTNAME>"
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: [FileSystem.renameSnapshot](#)

Delegation Token Operations

Get Delegation Token

- Submit a HTTP GET request.

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/?op=GETDELEGATIONTOKEN&renewer=<USER>&serv
```

The client receives a response with a [Token](#) JSON object:

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "Token":
  {
    "urlString": "JQAIAg9y..."
  }
}
```

See also: [renewer](#), [FileSystem.getDelegationToken](#), [kind](#), [service](#)

Get Delegation Tokens

- Submit a HTTP GET request.

```
curl -i "http://<HOST>:<PORT>/webhdfs/v1/?op=GETDELEGATIONTOKENS&renewer=<USER>"
```

The client receives a response with a [Tokens JSON object](#):

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{
  "Tokens":
  {
    "Token":
    [
      {
        "urlString": "KAAKSm9i ..."
      }
    ]
  }
}
```

See also: [renewer](#), [FileSystem.getDelegationTokens](#)

Renew Delegation Token

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/?op=RENEWDELEGATIONTOKEN&token=<TOK
```

The client receives a response with a [long JSON object](#):

```
HTTP/1.1 200 OK
Content-Type: application/json
Transfer-Encoding: chunked

{"long": 1320962673997}           //the new expiration time
```

See also: [token](#), [FileSystem.renewDelegationToken](#)

Cancel Delegation Token

- Submit a HTTP PUT request.

```
curl -i -X PUT "http://<HOST>:<PORT>/webhdfs/v1/?op=CANCELDELEGATIONTOKEN&token=<TOK
```

The client receives a response with zero content length:

```
HTTP/1.1 200 OK
Content-Length: 0
```

See also: `token`, `FileSystem.cancelDelegationToken`

Error Responses

When an operation fails, the server may throw an exception. The JSON schema of error responses is defined in [RemoteException JSON Schema](#). The table below shows the mapping from exceptions to HTTP response codes.

HTTP Response Codes

Exceptions	HTTP Response Codes
IllegalArgumentException	400 Bad Request
UnsupportedOperationException	400 Bad Request
SecurityException	401 Unauthorized
IOException	403 Forbidden
FileNotFoundException	404 Not Found
RumtimeException	500 Internal Server Error

Below are examples of exception responses.

Illegal Argument Exception

```
HTTP/1.1 400 Bad Request
Content-Type: application/json
Transfer-Encoding: chunked

{
  "RemoteException":
  {
    "exception"      : "IllegalArgumentException",
    "javaClassName" : "java.lang.IllegalArgumentException",
    "message"       : "Invalid value for webhdfs parameter \"permission\": ..."
  }
}
```

Security Exception

```
HTTP/1.1 401 Unauthorized
```

```
Content-Type: application/json
Transfer-Encoding: chunked
```

```
{
  "RemoteException":
  {
    "exception"      : "SecurityException",
    "javaClassName"  : "java.lang.SecurityException",
    "message"        : "Failed to obtain user group information: ..."
  }
}
```

Access Control Exception

```
HTTP/1.1 403 Forbidden
Content-Type: application/json
Transfer-Encoding: chunked
```

```
{
  "RemoteException":
  {
    "exception"      : "AccessControlException",
    "javaClassName"  : "org.apache.hadoop.security.AccessControlException",
    "message"        : "Permission denied: ..."
  }
}
```

File Not Found Exception

```
HTTP/1.1 404 Not Found
Content-Type: application/json
Transfer-Encoding: chunked
```

```
{
  "RemoteException":
  {
    "exception"      : "FileNotFoundException",
    "javaClassName"  : "java.io.FileNotFoundException",
    "message"        : "File does not exist: /foo/a.patch"
  }
}
```

JSON Schemas

All operations, except for [OPEN](#), either return a zero-length response or a JSON response. For [OPEN](#), the response is an octet-stream. The JSON schemas are shown below. See [draft-zyp-json-schema-03](#) for the syntax

definitions of the JSON schemas.

Note that the default value of `additionalProperties` is an empty schema which allows any value for additional properties. Therefore, all WebHDFS JSON responses allow any additional property. However, if additional properties are included in the responses, they are considered as optional properties in order to maintain compatibility.

ACL Status JSON Schema

```
{
  "name"      : "AclStatus",
  "properties":
  {
    "AclStatus":
    {
      "type"      : "object",
      "properties":
      {
        "entries":
        {
          "type": "array"
          "items":
          {
            "description": "ACL entry.",
            "type": "string"
          }
        },
        "group":
        {
          "description": "The group owner.",
          "type"      : "string",
          "required"   : true
        },
        "owner":
        {
          "description": "The user who is the owner.",
          "type"      : "string",
          "required"   : true
        },
        "stickyBit":
        {
          "description": "True if the sticky bit is on.",
          "type"      : "boolean",
          "required"   : true
        }
      }
    }
  }
}
```

XAttrs JSON Schema

```
{
  "name"      : "XAttrs",
```

```

"properties":
{
  "XAttrs":
  {
    "type"      : "array",
    "items":
    {
      "type"      : "object",
      "properties":
      {
        "name":
        {
          "description": "XAttr name.",
          "type"       : "string",
          "required"   : true
        },
        "value":
        {
          "description": "XAttr value.",
          "type"       : "string"
        }
      }
    }
  }
}

```

XAttrNames JSON Schema

```

{
  "name"      : "XAttrNames",
  "properties":
  {
    "XAttrNames":
    {
      "description": "XAttr names.",
      "type"       : "string",
      "required"   : true
    }
  }
}

```

Boolean JSON Schema

```

{
  "name"      : "boolean",
  "properties":
  {
    "boolean":
    {
      "description": "A boolean value",

```

```

        "type"      : "boolean",
        "required"  : true
      }
    }
  }
}

```

See also: [MKDIRS](#), [RENAME](#), [DELETE](#), [SETREPLICATION](#)

ContentSummary JSON Schema

```

{
  "name"      : "ContentSummary",
  "properties":
  {
    "ContentSummary":
    {
      "type"      : "object",
      "properties":
      {
        "directoryCount":
        {
          "description": "The number of directories.",
          "type"       : "integer",
          "required"   : true
        },
        "fileCount":
        {
          "description": "The number of files.",
          "type"       : "integer",
          "required"   : true
        },
        "length":
        {
          "description": "The number of bytes used by the content.",
          "type"       : "integer",
          "required"   : true
        },
        "quota":
        {
          "description": "The namespace quota of this directory.",
          "type"       : "integer",
          "required"   : true
        },
        "spaceConsumed":
        {
          "description": "The disk space consumed by the content.",
          "type"       : "integer",
          "required"   : true
        },
        "spaceQuota":
        {
          "description": "The disk space quota.",
          "type"       : "integer",
          "required"   : true
        }
      }
    }
  }
}

```

```
}
```

See also: [GETCONTENTSUMMARY](#)

FileChecksum JSON Schema

```
{
  "name"      : "FileChecksum",
  "properties":
  {
    "FileChecksum":
    {
      "type"      : "object",
      "properties":
      {
        "algorithm":
        {
          "description": "The name of the checksum algorithm.",
          "type"       : "string",
          "required"   : true
        },
        "bytes":
        {
          "description": "The byte sequence of the checksum in hexadecimal.",
          "type"       : "string",
          "required"   : true
        },
        "length":
        {
          "description": "The length of the bytes (not the length of the string).",
          "type"       : "integer",
          "required"   : true
        }
      }
    }
  }
}
```

FileStatus JSON Schema

```
{
  "name"      : "FileStatus",
  "properties":
  {
    "FileStatus": fileStatusProperties    //See FileStatus Properties
  }
}
```

See also: [FileStatus Properties](#), [GETFILESTATUS](#), [FileStatus](#)

FileStatus Properties

JavaScript syntax is used to define fileStatusProperties so that it can be referred in both FileStatus and FileStatuses JSON schemas.

```
var fileStatusProperties =
{
  "type"      : "object",
  "properties":
  {
    "accessTime":
    {
      "description": "The access time.",
      "type"       : "integer",
      "required"   : true
    },
    "blockSize":
    {
      "description": "The block size of a file.",
      "type"       : "integer",
      "required"   : true
    },
    "group":
    {
      "description": "The group owner.",
      "type"       : "string",
      "required"   : true
    },
    "length":
    {
      "description": "The number of bytes in a file.",
      "type"       : "integer",
      "required"   : true
    },
    "modificationTime":
    {
      "description": "The modification time.",
      "type"       : "integer",
      "required"   : true
    },
    "owner":
    {
      "description": "The user who is the owner.",
      "type"       : "string",
      "required"   : true
    },
    "pathSuffix":
    {
      "description": "The path suffix.",
      "type"       : "string",
      "required"   : true
    },
    "permission":
    {
      "description": "The permission represented as a octal string.",
      "type"       : "string",
      "required"   : true
    },
    "replication":
```

```

{
  "description": "The number of replication of a file.",
  "type"       : "integer",
  "required"   : true
},
"symlink":
{
  "description": "The link target of a symlink.",
  "type"       : "string"
},
"type":
{
  "description": "The type of the path object.",
  "enum"       : ["FILE", "DIRECTORY", "SYMLINK"],
  "required"   : true
}
}
};

```

FileStatuses JSON Schema

A FileStatuses JSON object represents an array of FileStatus JSON objects.

```

{
  "name"       : "FileStatuses",
  "properties":
  {
    "FileStatuses":
    {
      "type"       : "object",
      "properties":
      {
        "FileStatus":
        {
          "description": "An array of FileStatus",
          "type"       : "array",
          "items"      : fileStatusProperties //See FileStatus Properties
        }
      }
    }
  }
}

```

See also: [FileStatus Properties](#), [LISTSTATUS](#), [FileStatus](#)

Long JSON Schema

```

{
  "name"       : "long",
  "properties":
  {

```

```
"long":
{
  "description": "A long integer value",
  "type"       : "integer",
  "required"   : true
}
}
```

See also: [RENEWDELEGATIONTOKEN](#),

Path JSON Schema

```
{
  "name"       : "Path",
  "properties":
  {
    "Path":
    {
      "description": "The string representation a Path.",
      "type"       : "string",
      "required"   : true
    }
  }
}
```

See also: [GETHOMEDIRECTORY](#), [Path](#)

RemoteException JSON Schema

```
{
  "name"       : "RemoteException",
  "properties":
  {
    "RemoteException":
    {
      "type"       : "object",
      "properties":
      {
        "exception":
        {
          "description": "Name of the exception",
          "type"       : "string",
          "required"   : true
        },
        "message":
        {
          "description": "Exception message",
          "type"       : "string",
          "required"   : true
        }
      }
    }
  }
}
```

```

        "javaClassName": //an optional property
        {
            "description": "Java class name of the exception",
            "type"       : "string",
        }
    }
}

```

See also: [Error Responses](#)

Token JSON Schema

```

{
  "name"      : "Token",
  "properties":
  {
    "Token": tokenProperties //See Token Properties
  }
}

```

See also: [Token Properties](#), [GETDELEGATIONTOKEN](#), the note in [Delegation](#).

Token Properties

JavaScript syntax is used to define tokenProperties so that it can be referred in both Token and Tokens JSON schemas.

```

var tokenProperties =
{
  "type"      : "object",
  "properties":
  {
    "urlString":
    {
      "description": "A delegation token encoded as a URL safe string.",
      "type"       : "string",
      "required"   : true
    }
  }
}

```

Tokens JSON Schema

A Tokens JSON object represents an array of Token JSON objects.

```
{
  "name"      : "Tokens",
  "properties":
  {
    "Tokens":
    {
      "type"      : "object",
      "properties":
      {
        "Token":
        {
          "description": "An array of Token",
          "type"       : "array",
          "items"      : "Token": tokenProperties    //See Token Properties
        }
      }
    }
  }
}
```

See also: [Token Properties](#), [GETDELEGATIONTOKENS](#), the note in [Delegation](#).

HTTP Query Parameter Dictionary

ACL Spec

Name	<code>aclspec</code>
Description	The ACL spec included in ACL modification operations.
Type	String
Default Value	<empty>
Valid Values	See Permissions and HDFS .
Syntax	See Permissions and HDFS .

XAttr Name

Name	<code>xattr.name</code>
Description	The XAttr name of a file/directory.
Type	String
Default Value	<empty>
Valid Values	Any string prefixed with <code>user./trusted./system./security..</code>
Syntax	Any string prefixed with <code>user./trusted./system./security..</code>

XAttr Value

Name	<code>xattr.value</code>
Description	The XAttr value of a file/directory.
Type	String
Default Value	<empty>

Valid Values	An encoded value.
Syntax	Enclosed in double quotes or prefixed with 0x or 0s.

See also: [Extended Attributes](#)

XAttr set flag

Name	flag
Description	The XAttr set flag.
Type	String
Default Value	<empty>
Valid Values	CREATE,REPLACE.
Syntax	CREATE,REPLACE.

See also: [Extended Attributes](#)

XAttr value encoding

Name	encoding		
Description	The XAttr value encoding.		
Type	String		
Default Value	<empty>		
Valid Values	text	hex	base64
Syntax	text	hex	base64

See also: [Extended Attributes](#)

Access Time

Name	accesstime
Description	The access time of a file/directory.
Type	long
Default Value	-1 (means keeping it unchanged)
Valid Values	-1 or a timestamp
Syntax	Any integer.

See also: [SETTIMES](#)

Block Size

Name	blocksize
Description	The block size of a file.
Type	long
Default Value	Specified in the configuration.
Valid Values	> 0
Syntax	Any integer.

See also: [CREATE](#)

Buffer Size

Name	<code>buffersize</code>
Description	The size of the buffer used in transferring data.
Type	int
Default Value	Specified in the configuration.
Valid Values	> 0
Syntax	Any integer.

See also: [CREATE](#), [APPEND](#), [OPEN](#)

Create Parent

Name	<code>createparent</code>
Description	If the parent directories do not exist, should they be created?
Type	boolean
Default Value	false
Valid Values	true
Syntax	true

See also: [CREATESYMLINK](#)

Delegation

Name	<code>delegation</code>
Description	The delegation token used for authentication.
Type	String
Default Value	<empty>
Valid Values	An encoded token.
Syntax	See the note below.

Note that delegation tokens are encoded as a URL safe string; see `encodeToString()` and `decodeFromString(String)` in `org.apache.hadoop.security.token.Token` for the details of the encoding.

See also: [Authentication](#)

Destination

Name	<code>destination</code>
Description	The destination path.
Type	Path
Default Value	<empty> (an invalid path)
Valid Values	An absolute FileSystem path without scheme and authority.
Syntax	Any path.

See also: [CREATESYMLINK](#), [RENAME](#)

Do As

Name	<code>doas</code>
Description	Allowing a proxy user to do as another user.
Type	String
Default Value	null
Valid Values	Any valid username.
Syntax	Any string.

See also: [Proxy Users](#)

Fs Action

Name	<code>fsaction</code>
Description	File system operation read/write/execute
Type	String
Default Value	null (an invalid value)
Valid Values	Strings matching regex pattern "[rwx-]{3}"
Syntax	"[rwx-]{3}"

See also: [CHECKACCESS](#),

Group

Name	<code>group</code>
Description	The name of a group.
Type	String
Default Value	<empty> (means keeping it unchanged)
Valid Values	Any valid group name.
Syntax	Any string.

See also: [SETOWNER](#)

Length

Name	<code>length</code>
Description	The number of bytes to be processed.
Type	long
Default Value	null (means the entire file)
Valid Values	>= 0 or null
Syntax	Any integer.

See also: [OPEN](#)

Modification Time

Name	<code>modificationtime</code>
Description	The modification time of a file/directory.
Type	long
Default Value	-1 (means keeping it unchanged)
Valid Values	-1 or a timestamp
Syntax	Any integer.

See also: [SETTIMES](#)

New Length

Name	<code>newlength</code>
Description	The size the file is to be truncated to.
Type	long
Valid Values	≥ 0
Syntax	Any long.

Offset

Name	<code>offset</code>
Description	The starting byte position.
Type	long
Default Value	0
Valid Values	≥ 0
Syntax	Any integer.

See also: [OPEN](#)

Old Snapshot Name

Name	<code>oldsnapshotname</code>
Description	The old name of the snapshot to be renamed.
Type	String
Default Value	null
Valid Values	An existing snapshot name.
Syntax	Any string.

See also: [RENAMESNAPSHOT](#)

Op

Name	<code>op</code>
Description	The name of the operation to be executed.
Type	enum
Default Value	null (an invalid value)
Valid Values	Any valid operation name.
Syntax	Any string.

See also: [Operations](#)

Overwrite

Name	<code>overwrite</code>
Description	If a file already exists, should it be overwritten?
Type	boolean
Default Value	false

Valid Values	true
Syntax	true

See also: [CREATE](#)

Owner

Name	<code>owner</code>
Description	The username who is the owner of a file/directory.
Type	String
Default Value	<empty> (means keeping it unchanged)
Valid Values	Any valid username.
Syntax	Any string.

See also: [SETOWNER](#)

Permission

Name	<code>permission</code>
Description	The permission of a file/directory.
Type	Octal
Default Value	755
Valid Values	0 - 1777
Syntax	Any radix-8 integer (leading zeros may be omitted.)

See also: [CREATE](#), [MKDIRS](#), [SETPERMISSION](#)

Recursive

Name	<code>recursive</code>
Description	Should the operation act on the content in the subdirectories?
Type	boolean
Default Value	false
Valid Values	true
Syntax	true

See also: [RENAME](#)

Renewer

Name	<code>renewer</code>
Description	The username of the renewer of a delegation token.
Type	String
Default Value	<empty> (means the current user)
Valid Values	Any valid username.
Syntax	Any string.

See also: [GETDELEGATIONTOKEN](#), [GETDELEGATIONTOKENS](#)

Replication

Name	<code>replication</code>
Description	The number of replications of a file.
Type	short
Default Value	Specified in the configuration.
Valid Values	> 0
Syntax	Any integer.

See also: `CREATE`, `SETREPLICATION`

Snapshot Name

Name	<code>snapshotname</code>
Description	The name of the snapshot to be created/deleted. Or the new name for snapshot rename.
Type	String
Default Value	null
Valid Values	Any valid snapshot name.
Syntax	Any string.

See also: `CREATESNAPSHOT`, `DELETESNAPSHOT`, `RENAMESNAPSHOT`

Sources

Name	<code>sources</code>
Description	A list of source paths.
Type	String
Default Value	<empty>
Valid Values	A list of comma seperated absolute FileSystem paths without scheme and authority.
Syntax	Any string.

See also: `CONCAT`

Token

Name	<code>token</code>
Description	The delegation token used for the operation.
Type	String
Default Value	<empty>
Valid Values	An encoded token.
Syntax	See the note in Delegation .

See also: `RENEWDELEGATIONTOKEN`, `CANCELDELEGATIONTOKEN`

Token Kind

Name	<code>kind</code>
Description	The kind of the delegation token requested
Type	String

Default Value	<empty> (Server sets the default kind for the service)
Valid Values	A string that represents token kind e.g "HDFS_DELEGATION_TOKEN" or "WEBHDFS delegation"
Syntax	Any string.

See also: [GETDELEGATIONTOKEN](#)

Token Service

Name	<code>service</code>
Description	The name of the service where the token is supposed to be used, e.g. ip:port of the namenode
Type	String
Default Value	<empty>
Valid Values	ip:port in string format or logical name of the service
Syntax	Any string.

See also: [GETDELEGATIONTOKEN](#)

Username

Name	<code>user.name</code>
Description	The authenticated user; see Authentication .
Type	String
Default Value	null
Valid Values	Any valid username.
Syntax	Any string.

See also: [Authentication](#)